

Our Environment

Science • Chapter 13 • Chapter Summary & Study Guide for Daksh

Every plastic bag, every left-over banana peel, every drop of water — they all return to nature, but in very different ways. This chapter explores how the **environment** works as an interconnected system of living and non-living components, the role of food chains and food webs, why some materials are biodegradable and others aren't, and the global concerns around the **ozone layer** and waste management.

Ecosystem — the basic unit

An **ecosystem** consists of all organisms in an area together with their non-living environment (temperature, water, soil, air). Components:

- **Abiotic** — non-living: light, temperature, soil, water, minerals, air.
- **Biotic** — living: producers, consumers, decomposers.
- Ecosystems may be *natural* (forest, lake, ocean) or *artificial / human-made* (crop field, garden, aquarium).

Producers, consumers, decomposers and food chains

Energy enters every ecosystem through the sun. **Producers** (green plants) use it to make food. **Consumers** depend on others — herbivores (cows, deer) eat plants; carnivores (lions, tigers) eat herbivores; omnivores (humans, bears) eat both. **Decomposers** (bacteria, fungi) break down dead remains, returning the nutrients to the soil and air.

Linear FOOD CHAIN (a sequence of who-eats-whom):
Sun → Grass (producer) → Grasshopper → Frog → Snake → Hawk
T1 T2 T3 T4 T5
(Trophic levels)

10% LAW of energy flow:
Only ~10% of energy at one trophic level is passed to the next.
Hence food chains rarely have more than 4 or 5 levels.

FOOD WEB = many interconnected food chains in an ecosystem.

The flow of energy is **unidirectional** (sun → producer → consumer → lost as heat); but the flow of **materials** (carbon, nitrogen, water) is **cyclic** — they keep getting recycled in nature.

Biological magnification

Toxic, non-biodegradable chemicals (pesticides like DDT, heavy metals like mercury) entering the food chain get progressively more concentrated at higher trophic levels — this is **biological magnification**. So a top predator (or human eating fish) may end up with the highest dose, even though the original water might have very little of the toxin.

Biodegradable vs non-biodegradable wastes

- **Biodegradable** — broken down by natural agents (microbes, enzymes) over short time. Examples: vegetable peels, paper, cotton clothes, wood, animal dung.
- **Non-biodegradable** — persist for long periods, accumulate. Examples: plastics, polythene bags, glass, aluminium cans, DDT, radioactive wastes.

We can manage waste better through the **4 Rs**: **Refuse** (avoid using harmful items), **Reduce** (consume less), **Reuse** (use a container again), **Recycle** (turn waste into new product). Composting and segregating waste at source are simple steps every household can take.

The ozone layer — its role and depletion

The **ozone (O₃)** layer in the upper atmosphere shields life on Earth from the sun's harmful **ultraviolet (UV)** radiation. UV radiation can cause skin cancer, cataracts and damage to plants and phytoplankton. In the

1980s, scientists discovered an ozone hole over Antarctica — caused mainly by **chlorofluorocarbons (CFCs)** from refrigerators, air-conditioners and aerosol sprays. The Montreal Protocol (1987) was an international agreement to phase out CFCs, and ozone levels have begun to recover slowly.

★ MUST-REMEMBER POINTS

- Ecosystem = abiotic + biotic components.
- Producers (green plants) → consumers (herbivore/carnivore/omnivore) → decomposers.
- Energy flow is unidirectional; matter flow is cyclic.
- 10% law: only 10% of energy passes to the next trophic level → food chains short.
- Biological magnification: toxins get concentrated up the food chain (DDT, Hg).
- Biodegradable (peels, paper) vs non-biodegradable (plastic, glass).
- Ozone layer protects from UV; CFCs cause depletion → Montreal Protocol (1987).

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